Sathvik Reddy Bhavanam

cs23btech11056@iith.ac.in | sathvikreddy853@gmail.com | +91 9108514471 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.18
Sri Chaitanya Junior College	Class XII (TSBIE)	2023	98.70%
Sri Kumaran Children's Home	Class X (CBSE)	2021	92.80%

SCHOLASTIC ACHIEVEMENTS

The Integral Cup: Ranked 296 nationally in The Integral Cup, a national-level mathematics competition.

NTSE Scholar: Qualified NTSE Stage-2 and received the scholarship among 8 - 9 lakh students.

EXPERIENCE

Databricks - Compute Platform, *Software Engineering Intern*

- Built a **fault injection** framework for serverful compute to **simulate cloud infrastructure failures** during cluster launch, including **VM launch latency** and **VM type stockouts**, and **zone stockouts**.
- Added **probes** to production code for AWS, GCP, and Azure that periodically test these failure scenarios across the cluster launch flow.

Prof. Saketha Nath, *Undergraduate Research Assistant*

- Conducting a literature survey on diffusion models, sampling techniques, flow matching, and rectified flows.

Prof. Shashank Vatedka, *Undergraduate Research Assistant*

- Worked on distributed mean estimation with correlated quantization. Currently working on quantized matrix multiplication.

SKILLS

CP Profiles: Codeforces (Specialist, Max Rating: 1403), Leetcode (Rating: 1650, Over 350 problems solved)

Zetamac Score: 61 (Highest)

Prog. Languages: C, C++, Rust, Python, RISC-V, SQL, MATLAB

Libraries/Frameworks: NumPy, Pytorch, Spark, Matplotlib, CVXPY, React, Spark, Arrow

Tools: GDB, Git, LaTex, Make, Bazel, Valgrind, Protobuf, k8s, Terraform

RELEVANT COURSES

Systems: Data Structures, Algorithms, Computer Architecture, Operating Systems, Database, Systems, Compilers, Computer Networks, Machine Learning Systems

ML: Foundations of ML, Deep Learning, Reinforcement Learning, Convex Optimization: Theory and Algorithms, Statistical Learning Theory, Advanced Stochastic Processes

Theoretical CS & Maths: Discrete Maths, Calculus, Differntial Eqns, Probability, Linear Algebra, Algorithms, Theory of Computation, Computational Number Theory, Quantum Computing, Pseudorandomness, Approximation Algorithms

PROJECTS

LLVM (Flang) - OS Contrib - PAUSE Keyword Support: (GitHub)

- Implemented support for the PAUSE keyword with numeric and string arguments in the Flang codegen, contributing to the LLVM open-source compiler project.

Transformer-Aware Warp Scheduling for GPGPUs (WIP): *Course Project under Prof. Abhijit Das* (GitHub)

- Developing a transformer-aware warp scheduling mechanism for GPGPUs, targeting improved execution efficiency for transformer workloads.

Shell Interface Description Language (WIP): (GitHub)

- Designed an interface for **invoking shell commands directly from programming languages**, representing command outputs as **structured in-memory data**.
- Built **automatic language-specific stub generation** to expose shell command outputs to the host language, analogous to **Protocol Buffers for shell commands**.

Heisen: *Course Project under Prof. Ramakrishna Upadrastha* (GitHub)

- Designed a **quantum programming language** with **reference-based variables** enforcing no-cloning semantics.
- Developed abstractions for **tensoring and decomposing quantum states and gates**, with **LLVM QIR generation**.

Network Fairness: *Course Project under Prof. Kotaro Kataoka* (GitHub)

- Designed an **eBPF-based system** for **per-cgroup bandwidth limiting** using **packet filtering and rate limiting**.
- Developed a **C-based userspace controller** for configuring per-cgroup limits.

GemOS - Memory Management syscalls: *Course Project under Prof. Saurabh Kumar* (GitHub)

- Implemented mmap, munmap, and mprotect with VMA creation, splitting, merging, and protection management.
- Implemented page-fault handling and **lazy page allocation** using 4-level page tables.
- Implemented **Copy-on-Write cfork()** with shared physical pages and reference counting.

RISC-V Simulator: *Course Project under Prof. Rajesh Kedia* (GitHub)

- Developed a **C++ RISC-V simulator** for step-by-step execution and inspection of registers and memory. Implemented **cache simulation** with various replacement policies (LRU, FIFO, Random), supporting different cache sizes and associativities.
- Built a custom **lexer, parser, and argument parser** with an interactive query-response interface. Selected for peer and faculty presentation.

EXTRACURRICULAR

Swimming - Inter IIT Aquatics: Represented IIT Hyderabad in Inter IIT Aquatics and secured **4th place in the relay**.

Marathon Running: Completed a **10 km marathon in 1 hour 5 minutes** (best time).